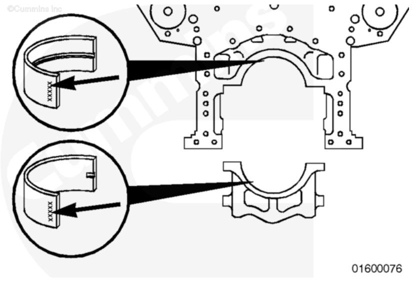
Trainings

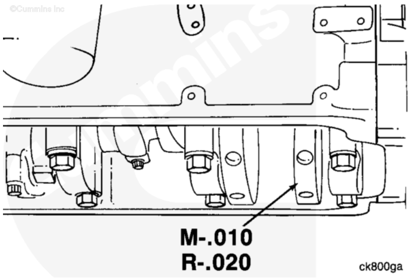
General Information

The upper bearings contain an oil hole. The lower bearings do **not** have an oil hole. Both bearings are marked on the back indicating location (upper or lower) and size (standard [STD] or oversize [OS]). The amount the bearing is oversize is indicated in inches.

Use the same size bearing; standard, 0.254 mm [0.010 in], 0.508 mm [0.020 in], or 0.762 mm [0.030 in] that was removed.



The crankshaft will be stamped on the end of the number one counterweight to indicate whether it has been ground undersize. The amounts of material removed from the main journals and the rod journals are both stamped at this location. Thrust bearing size is stamped on a crankshaft counterweight adjacent to the thrust location.



Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ **WARNING** ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Reference the original equipment manufacturer service information.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
- Remove the lubricating oil filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html)
- Remove the lubricating oil pan adapter assembly. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html)
- Remove the oil suction tube and brackets. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-035-tr.html)
- Remove the lubricating oil pump. Refer to Procedure 007-031 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-031-tr.html)

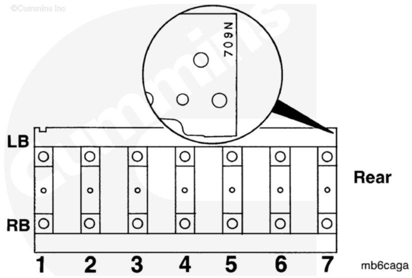
Remove

Check to make sure all of the main bearing caps are identified correctly.

The cylinder block has a number that is used to keep the block and main bearing caps together. The number is stamped on the bottom (oil pan flange) of the block at the left bank rear of the engine.

Note : "709N" is used as an example only.

Note : The K38 and QSK38 main bearing caps are shown in this illustration. The K50 and QSK50 main bearing caps are numbered 1 through 9.



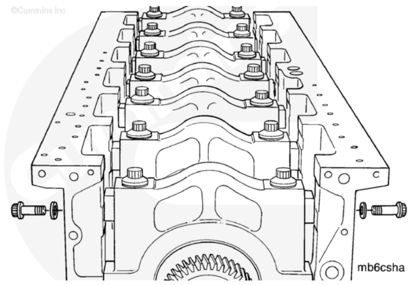
⚠ CAUTION ⚠

Personal injury and damage to the crankshaft can result if the crankshaft is dropped.

Note : Access to some side capscrews involves the removal of other components.

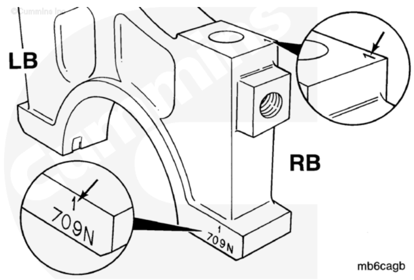
There are two side bolts for each main bearing cap.

Remove the main bearings one at a time.



Each main bearing cap contains the same number that is stamped on the block. The cap's location number is on the cap in two places.

If the cap is **not** marked for location, it is strongly recommended that it be marked before removal from the cylinder block to aid identification.



Remove the main bearing cap mounting capscrews.

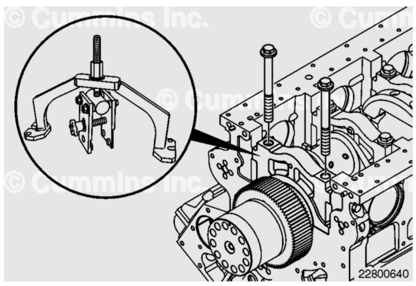
Remove the cap., Use the main bearing cap puller, Part Number 4919357.

Align the main bearing cap puller with the main bearing cap.

Install the fixture to the left and right oil pan rails of the engine block.

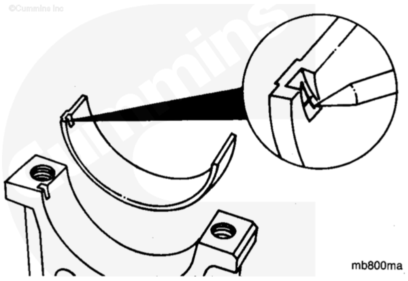
Install the main bearing cap puller clamp plate onto the main bearing cap and tighten the clamp knob.

Turn the adjusting nut at the top of the fixture with a wrench to lift the main bearing cap.



Note : If the flywheel housing is still in place, the rear main bearing cap will need to be tilted forward slightly for tool access. Use a pry bar or a large screwdriver to tilt the cap forward.

Remove the bearing from the cap and mark the location in the tang for future identification or analysis.

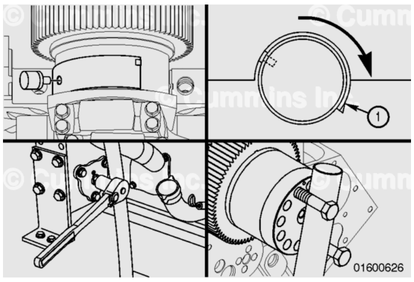


To remove the upper main bearing shells, use the main bearing remover, Part Number 3823818.

Rotate the crankshaft so the tang (1) of the main bearing shell rolls out of the block first.

Use the barring device to slowly rotate the crankshaft. Continue rotating the crankshaft until the bearing shell is out of the engine block. If the flywheel housing has been removed, the crankshaft can be turned using the crankshaft adapter mounting capscrews.

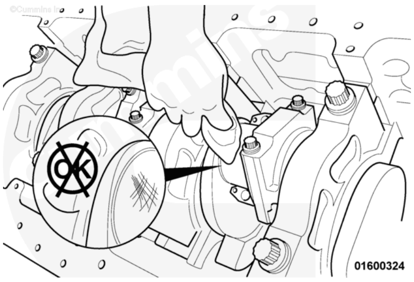
Mark the upper main bearing location in the tang area of the bearing.



Note : If bearing replacement was initiated due to damage, check the oil pan for debris. If significant debris is present, remove and disassemble the lubricating oil pump and inspect the bushing for debris damage. Refer to Procedure 007-031 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-031-tr.html)

Clean the crankshaft journal with a lint-free cloth.

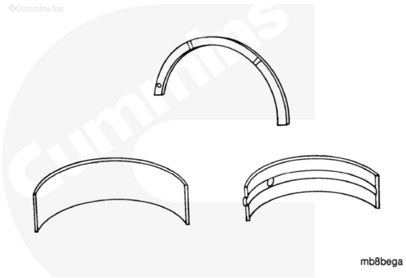
Inspect the journal for damage.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacture's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

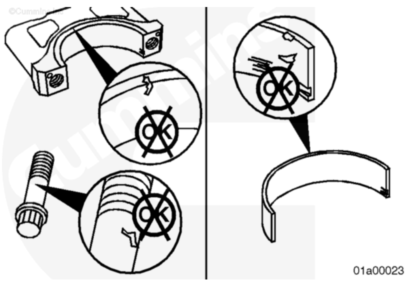
Do not use a scraper or a wire brush, this can damage the bearings.

Make sure the bearings are marked for location. The bearing **must** be installed in its original location if it is to be used again.

Use solvent to clean the bearings.

Dry with compressed air.

Inspect the main bearing caps, capscrews, and washers for the following damage:

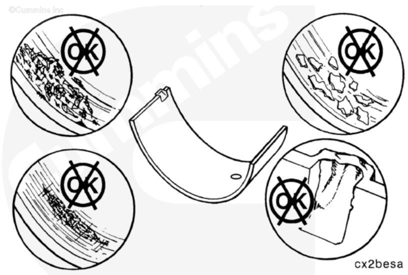


- Pitting
- Cracking
- Thread damage.

Replace any damaged part.

Inspect the head of each capscrew to count the number of punch marks. Any capscrews with five punch marks on the head **must** be replaced.

Inspect the bearings for damage.

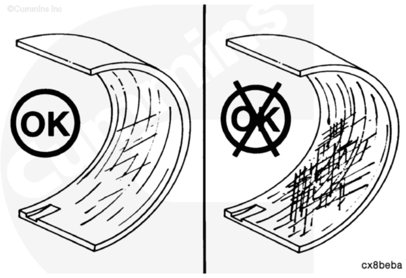


Replace any bearings that have the following damage:

- Nicks
- Cracks
- Burrs
- Scratches (deep enough to be felt with a fingernail)
- Scrapes
- Gouges
- Embedded particles

- Fretting
- Pitting
- Flaking
- Corrosion
- Lock tang damage.

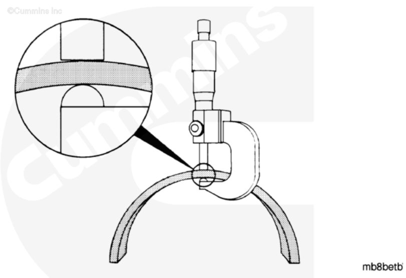
Normal bearing wear produces a smooth finish, which will wear into the copper lining. Exposed copper does **not always** indicate worn bearings. Reference the Parts Reuse Guidelines, Bulletin 3810303.



If large areas of copper lining are visible in the bearings before the engine has accumulated 3000 hours, inspect the engine for contamination from fine dirt particles and correct the problem.

For more detailed information of bearing damage, reference the Analysis and Prevention of Bearing Failures Manual, Bulletin 3810387.

Use a ball-end micrometer. Measure the bearing thickness at the wear location. The bearing **must** be replaced if it is **not** within specifications.



Main Bearing Thickness Standard or Oversize (OS)			
	mm		in
Standard	4.280	MIN	0.1685
	4.336	MAX	0.1707
0.010 (OS)	4.407	MIN	0.1735
	4.463	MAX	0.1757
0.020 (OS)	4.534	MIN	0.1785
	4.590	MAX	0.1807
0.030 (OS)	4.661	MIN	0.1835
	4.717	MAX	0.1857
0.040 (OS)	4.788	MIN	0.1885
	4.844	MAX	0.1907

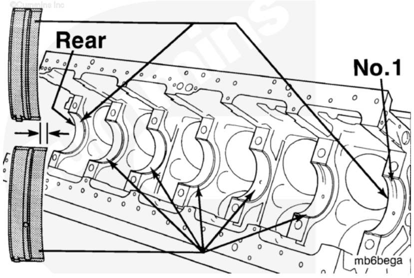
Install

Note : The location numbers of the main bearings begin with Number 1 at the front of the cylinder block.

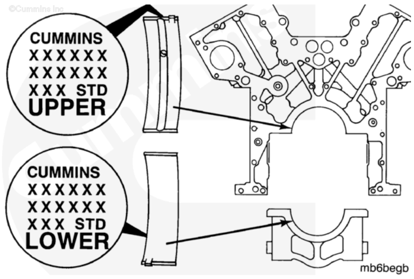
Note : K38 bearings are numbered 1 through 7.

Note : K50 bearings are numbered 1 through 9.

Main bearings are two widths. The narrow bearings fit locations Number 1 and Number 7 in the K38 block, and Number 1 and Number 9 in the K50 and QSK50 block. The wide bearings fit the remaining locations in both blocks.

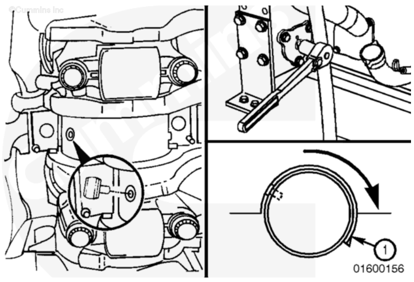


The upper bearings contain an oil hole. The lower bearings do **not**. Both bearings are marked on the back to indicate the location and either standard (STD) or oversize (OS). The amount of OS is stamped on the back in inches.



⚠ CAUTION ⚠

Prevent dirt from mixing with the lubricant. Dirty lubricant can cause low mileage failures.



Use a lint-free cloth. Clean the bearing and the mounting surface.

Note : Do not lubricate the back of the bearing.

Align the tang in the bearing with the slot in the block. Install the bearings. The end of the bearing **must** be even with the mounting surface for the main bearing cap.

Use engine oil if the engine will be operated immediately. Use Lubriplate™ 105, or equivalent, if the engine will **not** be operated immediately. Lubricate the bearings.

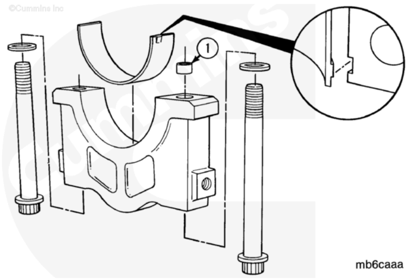
Reused bearings **mus**t be installed in their original location.

Install the upper main bearings shells. Use the same method that was used for the removal of the shells. Locate the rollout tool on the tang end of the bearing.

The bearing tang **must** fit into the slot in the bearing saddle to secure the bearing in the proper location.

⚠ CAUTION ⚠

Only newer K38 and K50 engines have the dowel ring (1) that protrudes from the main bearing cap. Do not install a cap with a dowel ring in the block that does not have a ring dowel on the other caps. If you are not sure, try to fit the dowel ring in the correct capscrew hole in the block. If it requires force, do not install the ring dowel.



If required, install the dowel ring.

Use a lint-free cloth to clean the lower bearings and the mounting surfaces of the cap.

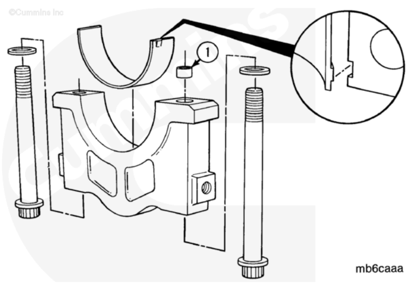
Align the tang in the bearing with the slot in the main bearing cap. The end of the bearing **must** be even with the mounting surface of the main bearing cap.

Use the same lubricant that was applied on the upper main bearings. Lubricate the lower bearings.

Note : Do not lubricate the backs of the main bearings.

⚠ CAUTION ⚠

Allow the excess oil to drip from the capscrews before installation in the block. Excess oil can cause hydraulic pressure, causing damage to the cylinder block.

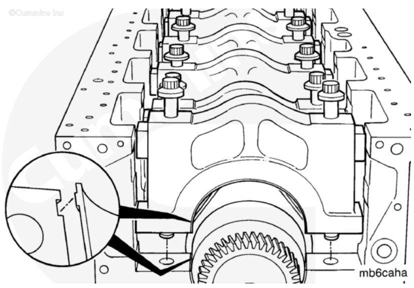


Use clean engine oil to lubricate the capscrew heads, threads, and washers.

Install the washers and capscrews in the main bearing cap.

⚠ CAUTION ⚠

Do not rotate the crankshaft until the caps are pulled to the block. Damage can result if the bearings move out-of-location.



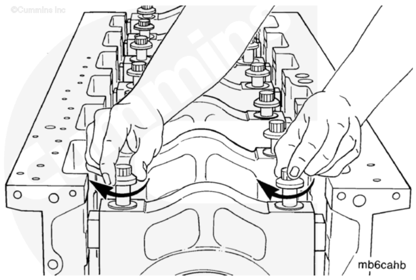
Note : Make sure the side of the cap and bearing with the bearing locating tang is toward the tang in the block.

Note : Keep the cap straight while moving it across the holes for the side bolts. There is little or no clearance between the parts.

Install the main bearing cap.

Note : Install the capscrews by hand.

Turn the capscrews until they touch the washer or the main bearing cap.

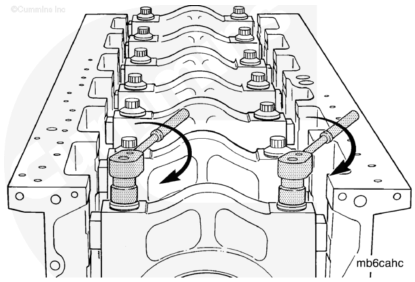


Note : Do not use impact wrenches. The main bearing shells can fall out.

Use both of the capscrews to pull the main bearing cap into position.

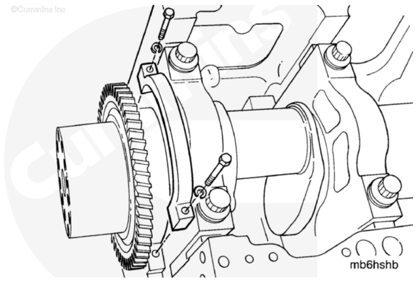
Use two wrenches. Tighten both of the capscrews at the same time.

Make sure the cap is touching the block. If it is **not**, check for a bearing out of location.



Note : The side of the lower thrust bearing support that is machined must be toward the main bearing cap.

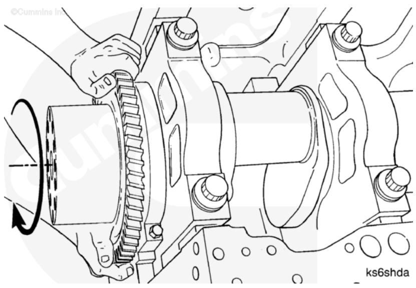
Install the lower crankshaft thrust bearing support, lock washers, and capscrews. Refer to Procedure 001-007 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-007-tr.html) Refer to Procedure 001-007 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-007-tr-gc.html)



⚠ CAUTION ⚠

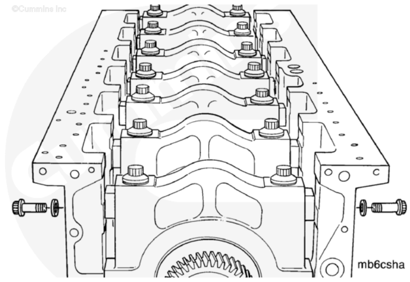
Do not rotate the crankshaft until the main bearing caps are pulled to the block. Damage will result if the bearings move out-of-location.

Turn the crankshaft with the barring tool. If it does **not** turn, remove the main bearing caps one at a time until the crankshaft turns freely. This helps to locate the cap or bearing that is **not** in alignment.



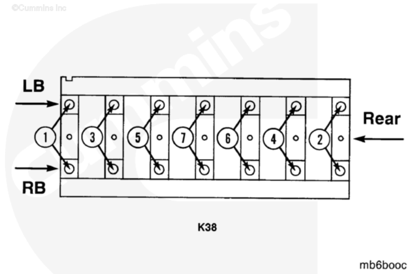
Use clean engine oil to lubricate the capscrew heads, threads, and hardened plain washers.

Install the washers and capscrews for the side of the main bearing cap. Each cap requires two washers and capscrews.



Note : The torque sequence reference is for the straight torque method only.

For K38 and QSK38 engines, use the following steps to tighten the capscrews in the sequence shown.

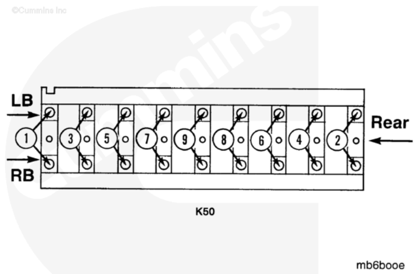


Torque Value:

1. 176 n•m [130 ft-lb]
2. 285 n•m [210 ft-lb]
3. 610 n•m [450 ft-lb]
4. Loosen
5. 176 n•m [130 ft-lb]
6. 285 n•m [210 ft-lb]
7. 610 n•m [450 ft-lb]

Note : The torque sequence reference is for the straight torque method only.

For K50 and QSK50 engines, use the following steps to tighten the capscrews in the sequence shown.

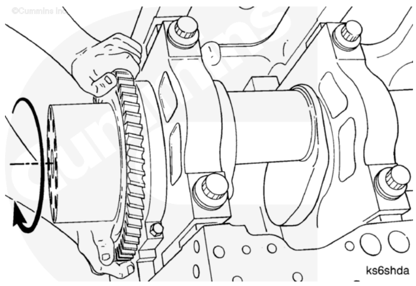


Torque Value:

1. 176 n•m [130 ft-lb]
2. 285 n•m [210 ft-lb]
3. 610 n•m [450 ft-lb]
4. Loosen
5. 176 n•m [130 ft-lb]
6. 285 n•m [210 ft-lb]
7. 610 n•m [450 ft-lb]

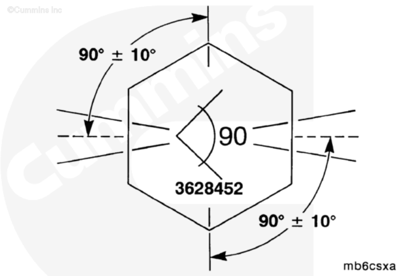
Turn the crankshaft with your hands. If it will **not** turn, loosen the main bearing caps one at a time until the crankshaft turns freely. This helps to locate the cap that is **not** in alignment.

Check to make sure there is end clearance.



Three different designs of capscrew have been used.

1. 12-point capscrews that are black
2. 6-point capscrews that are light gray
3. 6-point capscrews with angle marking (Part Number 3628452).



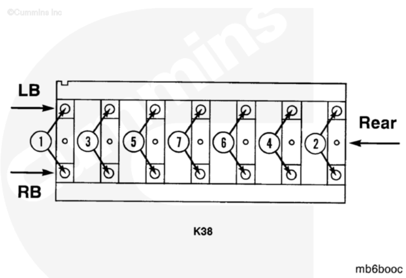
Note : Capscrews (1 and 2) use the torque procedure for tightening. They can be intermixed and installed in the same cap. Capscrew (3) uses the torque turn procedure for tightening.

Note : All engines built using the torque turn procedure are to have the torque turn procedure applied when rebuilt or repaired in the field. These engines can be identified by the capscrew markings. Older engines, unless block line bored, retain the old torque values. In cases where the torque turn procedure is not possible, due to limited access, a torque value of 814 N•m [600 ft-lb] can be used in place of the torque turn procedure.

Note : As the torque required for the torque turn procedure is significantly higher than the torque only procedure, Cummins Inc. recommends that a torque multiplier and breaker bar be used when advancing the capscrew the final 90 degrees. A standard torque wrench with 407 N•m [300 ft-lb] minimum capability, is to be used (without the torque multiplier) to obtain the initial 339 N•m [250 ft-lb] torque value.

Note : The below torque sequence is for the torque plus angle method.

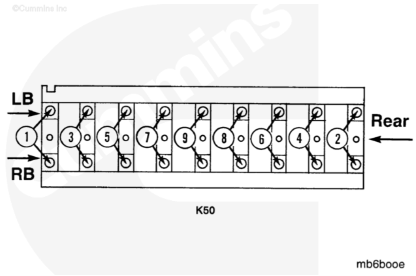
For K38 and QSK38 engines, tighten the main bearing cap mounting capscrews alternately and evenly, use the torque sequence shown.



Torque Value:

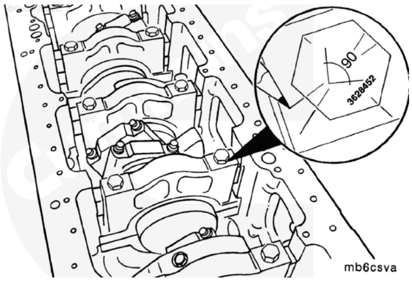
1. 176 n•m [130 ft-lb]
2. 339 n•m [250 ft-lb]
3. Loosen
4. 339 n•m [250 ft-lb]
5. +90 degrees

For K50 and QSK50 engines, tighten the main bearing cap mounting capscrews alternately and evenly. Use the torque sequence shown.



Note : The below torque sequence is for the torque plus angle method.

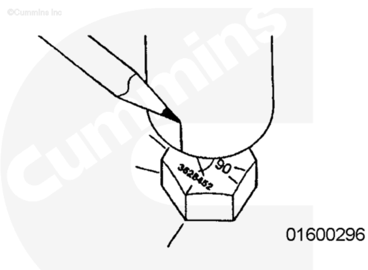
Use a pencil or scribe and mark three marks on the main bearing caps adjacent to three of the marks on the head of the capscrew as shown. Use Dykem™ on the cap surface, if necessary, to make the marks visible.



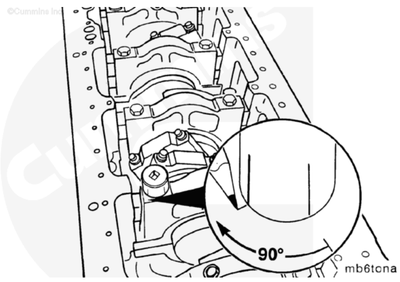
Torque Value:

1. 176 n•m [130 ft-lb]
2. 339 n•m [250 ft-lb]
3. Loosen
4. 339 n•m [250 ft-lb]
5. +90 degrees

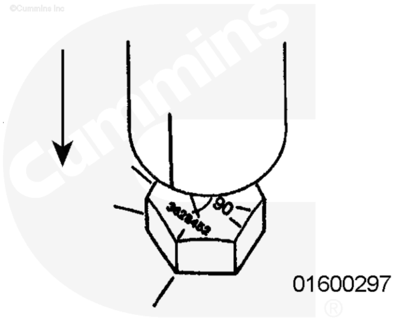
Install the socket on the capscrew and scribe a line on the socket, in line with the single line on the cylinder block. Use Dykem™ on the socket for clarity. This socket is to fit snugly over the capscrew head. Use of a worn socket will reduce accuracy.



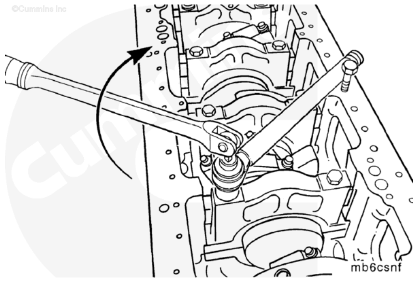
An alternative method for marking the cylinder block is to scribe the two lines 90 degrees apart on a socket. A template is to be used for accuracy. Place the socket over the capscrew and transfer one line to the cylinder block. Advance the capscrew 90 degrees. The second line on the socket needs to line up with the line on the block after the torque turn procedure is completed.



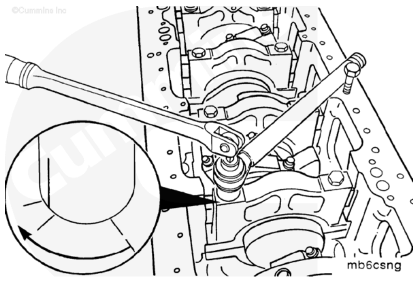
Place the socket over the capscrew with the mark in line with the single mark on the bearing cap.



Install a 7/16 inch capscrew, approximately 6 inches long, in the appropriate oil pan mounting capscrew hole to prevent the torque multiplier from turning when torque is applied to the main bearing capscrew.

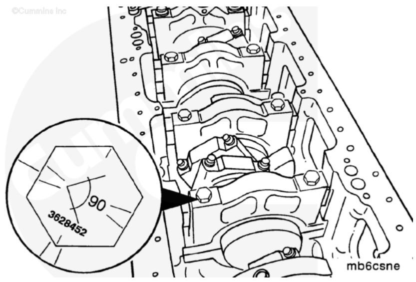


Use the torque multiplier, as shown, to tighten the capscrew until the line on the socket is midway between the two marks on the main bearing cap.



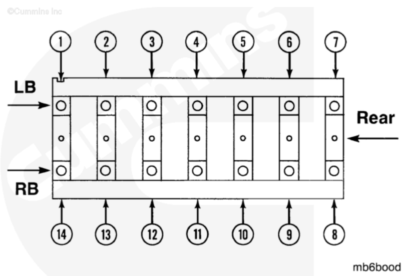
Shown is the proper final location for the capscrew head markings relative to the cap markings.

Repeat the procedure for the remaining capscrew. Use the correct torque sequence.



Note : Shown in this illustration is the K38 and QSK38. The K50 and QSK50 uses the same sequence, but is numbered 1 through 18.

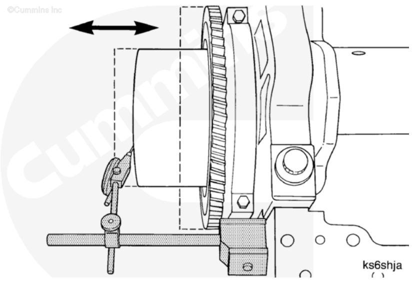
Use the following steps to manually tighten the side bolts in the sequence shown.



Torque Value:

1. 68 n•m [50 ft-lb]
2. 217 n•m [160 ft-lb]
3. 412 n•m [304 ft-lb]

Check the crankshaft end clearance. Refer to Procedure 001-007 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-007-tr.html)



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the lubricating oil pump. Refer to Procedure 007-031 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-031-tr.html)
- Install the oil suction tube and brackets. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-035-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
- Install the lubricating oil pan adapter assembly. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html)

- Install new lubricating oil filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html)
- Connect the batteries. Reference the original equipment manufacturer service information.
- Operate the engine and check for leaks.

Last Modified: 23-Oct-2023